

Vedant Borkute

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Education

University of California, Riverside

MS in Computer Science

Sep 2025 – Dec 2026

Riverside, California

Indian Institute of Technology, Bombay

B.Tech in Computer Science

Aug 2019 – Apr 2023

Mumbai, Maharashtra, India

Experience

Research Intern

CompFly AI

Jun 2026 – Sep 2026

San Francisco, CA

- Designed and built a reproducible benchmark harness for CompFly AI's **governance controls**, spanning AgentDojo prompt-injection attacks and ToolEmu high-stakes tool-use scenarios; operationalized expected-achievement criteria as action controls.
- Optimized a prompt-injection detection pipeline by benchmarking lightweight **transformer-based sequence classifiers**; tuned chunking, window/stride settings, heuristic rules, and parallel OR ensembles, then published findings in a technical blog.
- Extended and maintained **OpenTelemetry** instrumentation across agent orchestration, LLM inference, and tool-execution paths, enabling trace-level analysis of agent turns, tool calls, sub-agent handoffs, token usage, and cost.

Data Scientist

Finarb AI

Aug 2023 - Aug 2025

Kolkata, India

- Implemented and benchmarked statistical models (SARIMA, VAR, Facebook Prophet) and deep learning models (LSTMs, Temporal Convolutional Networks) for pharmaceutical supply chain demand forecasting, achieving a **90% reduction in Mean Absolute Percentage Error** over already present approaches (3MA, 6MA, SES).
- Developed an **end-to-end forecasting pipeline** encompassing data preprocessing, feature engineering, model selection with custom metrics, and Docker-based containerization for reproducible deployment across supply chain nodes.
- Built **AI agents** to automate analytical workflows:
 - Feature Engineering: domain-relevant features via RAG on internet/other sources (+5% model performance)
 - Cross-Database Queries: natural-language queries across databases via a SQL + Python code-generation agentic workflow
 - Business Dashboards: automated KPI dashboards with intuitive visualizations

Projects

AI for Navigation: Comparative Study of LLM Reasoning Strategies

Spring 2026

Advisor: Prof. Yinglun Zhu

- Co-authored a controlled study comparing five LLM agent strategies (Reactive, ReAct, Chain-of-Thought, Tree-of-Thought, Buffer-of-Thought) for sequential decision-making in MiniGrid navigation tasks.
- Designed a unified evaluation pipeline in Python around Qwen2.5-7B-Instruct with deterministic settings (20 episodes/environment, max 100 steps), and tracked success rate, exploration steps, and token usage.
- Built the Buffer-of-Thought agent with structured output and reusable strategy templates (direct navigation, obstacle avoidance, lava crossing, recovery), plus safety checks for unsafe forward actions.
- Demonstrated planning-performance trade-offs across increasing task difficulty: achieved up to **55% success** on LavaGapS6 and **20% success** on LavaCrossingS9N2, with Pareto analysis of accuracy vs. token cost.

Technical Skills

Languages: C++, R, JavaScript, Python, Bash, Prolog, AWK, MIPS, Neo4J CQL, SQL, Sed, Lex, Yacc, L^AT_EX

Technologies: Pandas, R, SQL, Power BI, NumPy, SciPy, MATLAB, PostgreSQL, React, Django, PyTorch, TensorFlow, MongoDB

Software: Microsoft Excel, RStudio, DataGrip, Cursor, Docker, Git, PowerPoint, Word, Slack, Azure DevOps, Jira, Confluence

Accomplishments

Ranked 1 in OBC-PwD in IIT-JEE Advanced 2019, India's most competitive exam, and among the top 0.8% nationwide in JEE Main among 1.3 million candidates.